

How To Select The Right WI-FI AND RF MODULES



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When it comes to throwing data over the air, there are many options. If you are confused about which wireless technology is best for your next project, then keep on reading.

Most standard, well-known types include transmitter modules, receiver modules, transceiver modules and system on a chip (SoC) modules. An SoC module is the same as a transceiver module, but it is often made with an onboard micro-controller (MCU). The MCU is typically used to handle radio data packetisation or managing a protocol such as an IEEE 802.15.4-compliant module. This type of module is typically used for designs that require additional processing for compliance with a protocol when the designer does not wish to incorporate this processing into the host MCU.

of the receiver also increases the effective communication range, but it also potentially causes malfunction due to interference from other RF devices.

RF modules selection

General RF. RF link devices are the most barebone radio communication devices with the cheapest solution. Both the receivers and transmitters are needed to make a complete one-way link. RF link of 434MHz is more common.

RF module devices are inexpensive as well as smarter, as these are sold as single-unit transceivers capable of both sending and receiving data. RFM22 is a low-cost ISM FSK transceiver module that offers communication at 433MHz ISM and adjustable output power of up to +20dBm.

Pololu Wixel is a general-purpose programmable module featuring 2.4GHz radio and USB. You can write your own software or load precompiled, open source apps onto TI CC2511F32 MCU at the heart of Wixel, turning it into a wireless serial link, data logger or whatever you need for your current project.

Bluetooth. Bluetooth wireless protocol has two versions: Bluetooth Low Energy, which is marketed as Bluetooth Smart, and classic Bluetooth, which is the original standard version of Bluetooth. Compared to classic Bluetooth, Bluetooth Smart is designed to address the needs of new wireless applications such as ultra-low power consumption, fast connection times, reliability and security.

Texas Instruments CC2540 and its improved version, CC2541, have been very popular for several years. Both devices feature an 8051 processor, which enables your application to be tied to 8051 processor.

Nordic Semiconductor nRF8001 is a standalone device that requires an external MCU with SPI interface. The benefit of using such a device is that you can use

Performance of a radio frequency (RF) module depends on a number of aspects. With increasing transmitter power, a larger communication distance can be achieved. However, this also results in a higher electrical power drain on the transmitter device, which causes shorter operating life for battery-powered devices.

Also, a higher transmitting power can make the program prone to interference from all other RF devices. Increasing the sensitivity

TABLE I
RF MODULES

Company	Description
Amazon	Nordic nRF24L01+ integrates a complete 2.4GHz RF transceiver, RF synthesiser and base-band logic including enhanced ShockBurst hardware protocol accelerator supporting a high-speed SPI interface for the application controller.
Jbtek	An overall Bluetooth module for Arduino board. If we factor in price, reliability in connection and data transfer as well as low power consumption, the module by Jbtek wins. You have the freedom of changing baud rate via AT commands from your computer while the board is compliant to Bluetooth V 2.0 standards.
Anaren	Zigbee/802.15.4 modules 2.4GHz internal antenna chip CC2530 radio
Digi International	Zigbee/802.15.4 modules XBee ZB S2C TH wire antenna